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Method of controlling power supply for amplifying radio frequency signal and communication apparatus for performing the same

Abstract

A method of controlling power supply for amplifying a radio frequency (RF) signal includes: determining whether a power supply mode for supplying power from a power modulator of a communication apparatus to an amplifier for amplifying the RF signal is an envelope tracking (ET) mode for supplying power based on ET of the RF signal; when the power supply mode is the ET mode, determining whether a downlink block error rate in the ET mode and a downlink modulation method in the ET mode satisfy a first condition; when the first condition is satisfied, determining whether a full resource block (RB) in an uplink band is allocated to uplink data transmission; and when the full RB in the uplink band is allocated to uplink data transmission, switching the power supply mode to an average power tracking (APT) mode for supplying power based on average power of the RF signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of International Application No. PCT/KR2022/016773 designating the United States, filed on Oct. 30, 2022, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2021-0157488, filed on Nov. 16, 2021, and Korean Patent Application No. 10-2022-0000231, filed on Jan. 3, 2022, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND

1. Field

(1) The disclosure relates to a method of controlling power supply for amplifying a radio frequency (RF) signal.

2. Description of Related Art

(2) A communication apparatus for transmitting and receiving a radio frequency (RF) signal may use an amplifier for amplifying the RF signal. Predetermined power may be supplied to the amplifier. To increase power efficiency, a method of supplying power to the amplifier may include an average power tracking (APT) method of supplying power based on average power of the RF signal and an envelope tracking (ET) method of supplying power based on ET of the RF signal.

SUMMARY

(3) When supplying power to an amplifier in an envelope tracking (ET) method to amplify a radio frequency (RF) signal, the possible occurrence of ET switching noise may decrease performance of receiving the RF signal by a communication apparatus. In addition, power consumption may increase when using an average power tracking (APT) mode.

(4) According to one example embodiment, a method of controlling power supply for amplifying an RF signal and a communication apparatus for performing the same may control an increase in power consumption when using an APT mode by setting a condition for switching between an ET mode and the APT mode.

(5) According to one example embodiment, a communication apparatus includes: an amplifier for amplifying an RF signal, a power modulator for supplying power to the amplifier, and a processor for controlling the power modulator. The processor is configured to determine whether a power supply mode for supplying the power from the power modulator to the amplifier is an ET mode for supplying the power based on ET of the RF signal, when the power supply mode is the ET mode, determine whether a downlink block error rate in the ET mode and a downlink modulation method in the ET mode satisfy a first condition, when the first condition is satisfied, determine whether a full resource block (RB) in an uplink band is allocated to uplink data transmission, and when the full RB in the uplink band is allocated to the uplink data transmission, switch the power supply mode to an APT mode for supplying the power based on average power of the RF signal.

(6) According to one example embodiment, a method of controlling power supply for amplifying an RF signal includes: determining whether a power supply mode for supplying power from a power modulator of a communication apparatus to an amplifier for amplifying the RF signal is an ET mode for supplying the power based on ET of the RF signal; when the power supply mode is the ET mode, determining whether a downlink block error rate in the ET mode and a downlink modulation method in the ET mode satisfy a first condition; when the first condition is satisfied, determining whether a full RB in an uplink band is allocated to uplink data transmission; and when

the full RB in the uplink band is allocated to the uplink data transmission, switching the power supply mode to an APT mode for supplying the power based on average power of the RF signal. (7) According to one example embodiment, a method of controlling power supply for amplifying an RF signal and a communication apparatus for performing the same may control an increase in power consumption when using an APT mode by setting a condition for switching between an ET mode and the APT mode.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a block diagram illustrating an electronic device in a network environment according to various example embodiments;

(3) FIG. 2 is a diagram illustrating envelope tracking (ET) switching noise caused when supplying power based on ET of a radio frequency (RF) signal in a communication apparatus according to one example embodiment;

(4) FIG. 3 is a block diagram illustrating a communication apparatus according to one example embodiment;

(5) FIGS. 4A and 4B are diagrams illustrating a condition on a downlink block error rate and a downlink modulation method in a power supply control method according to one example embodiment; and

(6) FIGS. 5 and 6 are flowcharts each illustrating a power supply control method according to one example embodiment.

DETAILED DESCRIPTION

(7) Hereinafter, example embodiments will be described in detail with reference to the accompanying drawings. When describing the example embodiments with reference to the accompanying drawings, like reference numerals refer to like elements and a repeated description related thereto will be omitted.

(8) FIG. 1 is a block diagram illustrating an electronic device **101** in a network environment **100** according to various example embodiments. Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or communicate with at least one of an electronic device **104** and a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an example embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an example embodiment, the electronic device **101** may include a processor **120**, a memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, and a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some example embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added to the electronic device **101**. In some example embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be integrated as a single component (e.g., the display module **160**).

(9) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101**

connected to the processor **120** and may perform various data processing or computations.

According to an example embodiment, as at least a part of data processing or computations, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in a volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in a non-volatile memory **134**. According to an example embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from or in conjunction with the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121** or to be specific to a specified function. The auxiliary processor **123** may be implemented separately from the main processor **121** or as a part of the main processor **121**.

(10) The auxiliary processor **123** may control at least some of functions or states related to at least one (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) of the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state or along with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an example embodiment, the auxiliary processor **123** (e.g., an ISP or a CP) may be implemented as a portion of another component (e.g., the camera module **180** or the communication module **190**) that is functionally related to the auxiliary processor **123**. According to an example embodiment, the auxiliary processor **123** (e.g., an NPU) may include a hardware structure specifically for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. The machine learning may be performed by, for example, the electronic device **101**, in which artificial intelligence is performed, or performed via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, for example, supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence (AI) model may include a plurality of artificial neural network layers. An artificial neural network may include, for example, a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), and a bidirectional recurrent deep neural network (BRDNN), a deep Q-network, or a combination of two or more thereof, but is not limited thereto. The AI model may additionally or alternatively include a software structure other than the hardware structure.

(11) The memory **130** may store various pieces of data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various pieces of data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

(12) The program **140** may be stored as software in the memory **130** and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

(13) The input module **150** may receive, from outside (e.g., a user) the electronic device **101**, a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

(14) The sound output module **155** may output a sound signal to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing a recording. The receiver may be used to receive an incoming call. According to an example embodiment, the receiver may be implemented separately from the speaker or as a part of the speaker.

(15) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a control circuit for controlling a display, a hologram device, or a projector and control circuitry to control its corresponding one of the display, the hologram device, and the projector. According to an example embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force of the touch.

(16) The audio module **170** may convert sound into an electric signal or vice versa. According to an example embodiment, the audio module **170** may obtain the sound via the input module **150** or output the sound via the sound output module **155** or an external electronic device (e.g., the electronic device **102**, such as a speaker or headphones) directly or wirelessly connected to the electronic device **101**.

(17) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101** and generate an electric signal or data value corresponding to the detected state. According to an example embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(18) The interface **177** may support one or more specified protocols to be used by the electronic device **101** to couple with the external electronic device (e.g., the electronic device **102**) directly (e.g., by wire) or wirelessly. According to an example embodiment, the interface **177** may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(19) The connecting terminal **178** may include a connector via which the electronic device **101** may physically connect to an external electronic device (e.g., the electronic device **102**). According to an example embodiment, the connecting terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphones connector).

(20) The haptic module **179** may convert an electric signal into a mechanical stimulus (e.g., a vibration or a movement) or an electrical stimulus, which may be recognized by a user via their tactile sensation or kinesthetic sensation. According to an example embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(21) The camera module **180** may capture a still image and moving images. According to an example embodiment, the camera module **180** may include one or more lenses, image sensors, ISPs, and flashes.

(22) The power management module **188** may manage power supplied to the electronic device **101**. According to an example embodiment, the power management module **188** may be implemented as, for example, at least a part of a power management integrated circuit (PMIC).

(23) The battery **189** may supply power to at least one component of the electronic device **101**. According to an example embodiment, the battery **189** may include, for example, a primary cell, which is not rechargeable, a secondary cell, which is rechargeable, or a fuel cell. The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more CPs that are operable independently from the processor **120** (e.g., an AP) and that support direct (e.g., wired) communication or wireless communication. According to an example embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication

module **194** (e.g., a local area network (LAN) communication module, or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device, for example, the electronic device **104**, via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., a LAN or a wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multiple components (e.g., multiple chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the SIM **196**.

(24) The wireless communication module **192** may support a 5G network after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., a mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (MIMO), full dimensional MIMO (FD-MIMO), an array antenna, analog beam-forming, or a large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**).

(25) According to an example embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(26) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an example embodiment, the antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an example embodiment, the antenna module **197** may include a plurality of antennas (e.g., an antenna array). In such a case, at least one antenna appropriate for a communication scheme used in a communication network, such as the first network **198** or the second network **199**, may be selected by, for example, the communication module **190** from the plurality of antennas. The signal or power may be transmitted or received between the communication module **190** and the external electronic device via the at least one selected antenna. According to an example embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as a part of the antenna module **197**.

(27) According to various example embodiments, the antenna module **197** may form a mmWave antenna module. According to an example embodiment, the mmWave antenna module may include a PCB, an RFIC on a first surface (e.g., the bottom surface) of the PCB, or adjacent to the first surface of the PCB and capable of supporting a designated high-frequency band (e.g., a mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the PCB, or adjacent to the second surface of the PCB and capable of transmitting or receiving signals of the designated high-frequency band.

(28) At least some of the above-described components may be coupled mutually and exchange

signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(29) According to an example embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device (e.g., the electronic device **104**) via the server **108** coupled with the second network **199**. Each of the external electronic devices (e.g., the electronic device **102** or **104**) may be a device of the same type as or a different type from the electronic device **101**. According to an example embodiment, all or some of operations to be executed by the electronic device **101** may be executed by one or more external electronic devices (e.g., the electronic devices **102** and **104** and the server **108**). For example, if the electronic device **101** needs to perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or service. The one or more external electronic devices receiving the request may perform the at least part of the function or service, or an additional function or an additional service related to the request and may transfer a result of the performance to the electronic device **101**. The electronic device **101** may provide the result, with or without further processing the result, as at least part of a response to the request. To that end, cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or MEC. In an example embodiment, the external electronic device (e.g., the electronic device **104**) may include an Internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an example embodiment, the external electronic device (e.g., the electronic device **104**) or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., a smart home, a smart city, a smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(30) FIG. 2 is a diagram illustrating envelope tracking (ET) switching noise caused when supplying power based on ET of a radio frequency (RF) signal in a communication apparatus (not shown) (e.g., the electronic device **101** of FIG. 1) according to one example embodiment.

(31) Referring to FIG. 2, illustrated are a power modulator **205** included in the communication apparatus (not shown) and an amplifier **215** for amplifying the RF signal by receiving power from the power modulator **205**. The RF signal amplified by the amplifier **215** may be a signal transmitted from the communication apparatus to the outside and may include data transmitted through a physical uplink shared channel (PUSCH).

(32) The communication apparatus (especially, the power modulator **205**) may supply power to the amplifier **215** in various power supply modes. For example, the communication apparatus (especially, the power modulator **205**) may supply predetermined voltage or supply power to the amplifier **215** in an ET mode, which is a power supply mode for supplying power based on ET of the RF signal, or in an average power tracking (APT) mode, which is a power supply mode for supplying power based on average power of the RF signal to increase power efficiency.

(33) Although supplying power in the ET mode is more power-efficient than supplying power in the APT mode, possible ET switching noise may affect reception performance of the communication apparatus.

(34) For example, in FIG. 2, the power modulator **205** may supply power to the amplifier **215** in the ET mode. The power modulator **205** may include an amplifier **210** for receiving and amplifying an envelope signal of an RF signal, a switch **220** that is on and off based on an output of the amplifier **210**, a capacitor C that is charged and discharged based on the on and off of the switch **220**, and an inductor L for supplying power charged and discharged in the capacitor C to the amplifier **215**.

(35) The envelope signal of an RF signal may be input to the power modulator **205**, and the input envelope signal may be amplified by the amplifier **210** of the power modulator **205**. The amplifier **210** may be a linear amplifier. The switch **220** may be on and off based on an output of the amplifier **210**, and output power of the power modulator **205** may be supplied to the amplifier **215** through the inductor L.

(36) Output voltage **225** of the power modulator **205** may be provided back as a differential input **230** of the amplifier **210**. A difference between the voltage of the envelope signal input to the amplifier **210** and the output voltage **225** of the power modulator **205** may be reflected, as an error, in the output of the amplifier **210**.

(37) The switch **220** may be repeatedly on and off based on the output of the amplifier **210** and may output voltage in a rectangular waveform. The rectangular-waveform voltage may cause ET switching noise.

(38) The ET switching noise may be applied to a reception port of an RFIC of the communication apparatus through front-end configuration, such as a duplexer, a low-noise amplifier (LNA), and a surface acoustic wave (SAW) filter of the communication apparatus, and may decrease the reception performance of the communication apparatus at maximum power.

(39) The ET switching noise may affect the reception performance more in a certain communication band than other bands. For example, the ET switching noise may affect the reception performance more in a communication band where an interval between an uplink band and a downlink band is relatively narrow. For example, when the communication apparatus uses B5 and B8 bands of long-term evolution (LTE) communication, the interval between the uplink and downlink bands is narrower than other LTE communication bands.

(40) Therefore, the ET switching noise may further affect the reception performance. In a communication band where the reception performance is degraded by the ET switching noise, the communication apparatus may supply power to the amplifier **215** by using the APT mode, but power efficiency may decrease.

(41) According to one example embodiment, a method of controlling power supply for amplifying an RF signal and a communication apparatus for performing the same may control an increase in power consumption when using an APT mode by setting a condition for switching between an ET mode and the APT mode.

(42) Hereinafter, referring to FIG. 3, descriptions are provided on the configuration of a method of controlling power supply for amplifying an RF signal and a communication apparatus for performing the same.

(43) FIG. 3 is a block diagram illustrating a communication apparatus according to one example embodiment.

(44) Referring to FIG. 3, a communication apparatus **300** may include an amplifier **320** for amplifying an RF signal transmitted to the outside through an antenna, a power modulator **315** for supplying power to the amplifier **320**, and a processor **305** for controlling the power modulator **315**. The processor **305** may include a communication processor for processing a digital signal.

(45) The communication apparatus **300** may process a transmitted or received RF signal and may include an RFIC **310** for performing conversion between analog and digital signals. The communication apparatus **300** may include a front-end circuit (not shown) for preprocessing an RF signal converted into an analog signal by the RFIC **310**, and the front-end circuit may include the amplifier **320**.

(46) The power modulator **315**, to reduce power loss in the amplifier **320**, may supply power to the amplifier **320** in an APT or ET mode. Referring to FIG. 3, a circuit of the power modulator **315** is illustrated as an example. For example, the power modulator **315** may include a first buck converter **325** for supplying power to the amplifier **320** in the APT mode. For example, the first buck converter **325** may be a buck boost converter.

(47) The power modulator **315** may include at least one linear amplifier for supplying power in the

ET mode. For example, the power modulator **315** may include a first linear amplifier **330** for converting power for 3-generation (3G) communication and LTE communication and a second linear amplifier **335** for converting power for 5G communication and 5G Sub-6 communication. To improve ET performance of a rapidly changing RF signal, the power modulator **315** may include a second buck converter **340**, which slowly operates compared to the first linear amplifier **330**. However, the configuration of the circuit of the power modulator **315** illustrated in FIG. 3 is merely an example, and the power modulator **315** may be variously configured.

(48) The amplifier **320** may include at least one amplifier based on the frequency feature of an RF signal to be amplified.

(49) The processor **305**, based on a downlink block error rate (BLER) and a downlink modulation method of an RF signal, may determine a power supply mode with respect to the amplifier **320** between any one of the APT mode and the ET mode, and based on the determined mode, may control the power modulator **315**.

(50) The processor **305** may determine whether the power supply mode for supplying power to the amplifier **320** from the power modulator **315** is the ET mode. The processor **305**, when the power supply mode is the ET mode, may determine whether a downlink BLER in the ET mode and a downlink modulation method in the ET mode satisfy a first condition. The processor **305**, when the first condition is satisfied, may determine whether a full resource block (RB) in an uplink band is allocated to uplink data transmission. The processor **305**, when the full RB in the uplink band is allocated to the uplink data transmission, may switch the power supply mode to the APT mode.

(51) The first condition may include a condition whether the downlink BLER in the ET mode is greater than or equal to a first reference downlink BLER corresponding to the downlink modulation method in the ET mode. For example, a downlink modulation method, such as quadrature phase shift keying (QPSK), 16-quadrature amplitude modulation (16-QAM), 64-QAM, and/or 256-QAM, may be used as the downlink modulation method in the ET mode for an RF signal, and the first condition may include a condition on the first reference downlink BLER corresponding to the downlink modulation method usable in the ET mode. When the downlink BLER in the ET mode corresponding to the downlink modulation method used in the ET mode is greater than or equal to the first reference downlink BLER, the processor **305** may determine that the first condition is satisfied.

(52) The first reference downlink BLER corresponding to the downlink modulation method in the ET mode in the first condition may be the same or differ depending on which downlink modulation method is used. For example, when the QPSK, 16-QAM, 64-QAM, or 256-QAM downlink modulation method is used in the ET mode, the first reference downlink BLER corresponding to the QPSK, 16-QAM, and 64-QAM downlink modulation methods in the first condition may be about 3%, and the first reference downlink BLER corresponding to the 256-QAM downlink modulation method may be about 5%. As another example, the first reference downlink BLER corresponding to the QPSK and 16-QAM downlink modulation methods may be about 3% and the first reference downlink BLER corresponding to the 64-QAM and 256-QAM downlink modulation methods may be about 5%. In yet another example, the first reference downlink BLER corresponding to the QPSK, 16-QAM, 64-QAM, and 256-QAM downlink modulation methods may be about 5%. As still another example, the first reference downlink BLER corresponding to the QPSK, 16-QAM, 64-QAM, and 256-QAM downlink modulation methods may be about 2%, 3%, 4%, and 5%, respectively.

(53) TABLE-US-00001

TABLE 1	First Condition	Downlink	First Reference	Modulation Method
Downlink BLER	QPSK 3% or more	16-QAM	64-QAM	256-QAM 5% or more

(54) Referring to Table 1, the first condition is provided as an example. A downlink BLER may vary depending on a downlink modulation method. For example, the downlink BLER may increase as the number of symbols used in downlink modulation increases. A symbol may be a unit representing the number of bits transmitted once in a modulation process. For example, 16-QAM

may modulate an RF signal by using 16 symbols, and 64-QAM may modulate an RF signal by using 64 symbols.

(55) The first reference downlink BLER corresponding to the downlink modulation method in the ET mode in the first condition may have a greater value as more symbols are used in the downlink modulation method in the ET mode. For example, in the example of Table 1, the downlink BLER may increase sequentially from QPSK to 16-QAM, 64-QAM, and 256-QAM.

(56) When using the QPSK, 16-QAM, 64-QAM, and 256-QAM as a downlink modulation method and applying the same first reference downlink BLER of about 3% to the QPSK, 16-QAM, 64-QAM, and 256-QAM, the 256-QAM downlink modulation method in the ET mode may frequently exceed the first reference downlink BLER of about 3% corresponding to the downlink BLER in the ET mode. Therefore, a greater value (e.g., 5%) than the first reference downlink BLER of the QPSK, 16-QAM, and 64-QAM downlink modulation methods may be desirable to be the first reference downlink BLER corresponding to the 256-QAM downlink modulation method.

(57) However, the downlink modulation methods provided herein and the first reference downlink BLER value corresponding to each downlink modulation method may only be an example for describing the first condition. In addition, the first condition may include a condition on various downlink modulation methods and a first reference downlink BLER on each downlink modulation method.

(58) The processor **305**, when the first condition is satisfied, may determine whether a full RB in an uplink band is allocated to uplink data transmission. The RB may refer to an allocable unit of frequency and may have a designated bandwidth. For example, the uplink band may be divided into a plurality of RBs and may be allocated to the uplink data transmission in RB units. For example, an uplink band with a 20 MHz bandwidth may be divided into 100 RBs, and the divided 100 RBs may be completely or partially allocated to the uplink data transmission.

(59) When a full RB in an uplink band is allocated to the uplink data transmission, ET switching noise occurring in the ET mode may affect more the reception performance of the communication apparatus **300** than when a partial RB in the uplink band is allocated to the uplink data transmission. For example, when the communication apparatus **300** uses a B5 band of LTE communication including an uplink band of about 824 MHz to 849 MHz and a downlink band of about 869 MHz to 894 MHz, when a full RB in the uplink band is allocated to the uplink data transmission, 849 MHz frequency, which is near the downlink band, may be used. Accordingly, ET switching noise may have a great impact on reception performance in the downlink band. When a partial RB in the uplink band is allocated to the uplink data transmission, a frequency of which the uplink band is relatively far from the downlink band may be used. Accordingly, ET switching noise may have a relatively small impact on the reception performance in the downlink band.

(60) The processor **305**, since the ET switching noise may have a great impact on the reception performance when the full RB in the uplink band is allocated to the uplink data transmission, may switch the power supply mode to the APT mode. For example, when the uplink band is 20 MHz, the processor **305** may switch the power supply mode to the APT mode when 100 RBs are completely allocated to the uplink data transmission.

(61) The processor **305**, when a partial RB, not the full RB, in the uplink band is allocated to the uplink data transmission, may compare a downlink BLER in the APT mode to a downlink

(62) BLER in the ET mode. The processor **305** may compare the downlink BLER in the APT mode with the downlink BLER in the ET mode through a physical uplink control channel (PUCCH).

(63) The processor **305**, when the downlink BLER in the APT mode is less than the downlink BLER in the ET mode, may switch the power supply mode to the APT mode.

(64) Based on the first condition, whether a full RB in an uplink band is allocated to the uplink data transmission, and/or a result of comparing a downlink BLER in the APT mode and a downlink BLER in the ET mode, the processor **305** may determine whether to switch from the ET mode to the APT mode. Accordingly, the processor **305** may reduce the number of times switching to the

APT mode, additional power consumed by using the APT mode, and deterioration of reception performance and downlink throughput (T-PUT) performance.

(65) To determine whether to switch the power supply mode from the APT mode to the ET mode, in the APT mode, the processor **305** may identify the downlink BLER in the ET mode and the downlink modulation method in the ET mode, and determine whether the downlink BLER in the ET mode and the downlink modulation method in the ET mode satisfy a second condition. The processor **305** may identify the downlink BLER in the ET mode and the downlink modulation method in the ET mode through the PUCCH.

(66) The second condition may include a condition whether the downlink BLER in the ET mode is less than a second reference downlink BLER corresponding to the downlink modulation method in the ET mode.

(67) The second condition may include a condition on a second reference downlink BLER corresponding to the downlink modulation method that may be used in the ET mode. The processor **305**, when the downlink BLER in the ET mode is less than the second reference downlink BLER corresponding to the downlink modulation method used in the ET mode, may determine that the second condition is satisfied.

(68) The second reference downlink BLER may be a downlink BLER less than the first reference downlink BLER by a threshold value (e.g., 2% or other designated values). The threshold value may be set to prevent frequent switching between the ET mode and the APT mode.

(69) TABLE-US-00002 TABLE 2 Second Condition Downlink Second Reference Modulation Method Downlink BLER QPSK less than 1% 16-QAM 64-QAM 256-QAM less than 3%

(70) Referring to Table 2, the second condition corresponding to the first condition in Table 1 is provided as an example. For example, when, in the first condition, the first downlink BLER corresponding to the QPSK, 16-QAM, and 64-QAM downlink modulation methods is about 3%, and the first reference downlink BLER corresponding to the 256-QAM downlink modulation method is about 5%, in the second condition, the second downlink BLER corresponding to the QPSK, 16-QAM, and 64-QAM downlink modulation methods may be about 1%, and the second reference downlink BLER corresponding to the 256-QAM downlink modulation method may be about 3%.

(71) As another example, when, in the first condition, the first downlink BLER corresponding to the QPSK and 16-QAM downlink modulation methods is around 3%, and the first reference downlink BLER corresponding to the 64-QAM and 256-QAM downlink modulation methods is about 5%, in the second condition, the second downlink BLER corresponding to the QPSK and 16-QAM downlink modulation methods may be about 1%, and the second reference downlink BLER corresponding to the 64-QAM and 256-QAM downlink modulation methods may be about 3%. In yet another example, when, in the first condition, the first reference downlink BLER corresponding to the QPSK, 16-QAM, 64-QAM, and 256-QAM downlink modulation methods is about 5%, in the second condition, the second reference downlink BLER corresponding to the QPSK, 16-QAM, 64-QAM, and 256-QAM downlink modulation methods may be about 3%. As still another example, when, in the first condition, the first reference downlink BLER corresponding to the QPSK, 16-QAM, 64-QAM, and 256-QAM downlink modulation methods is about 2%, 3%, 4%, and 5%, respectively, in the second condition, the second reference downlink BLER corresponding to the QPSK, 16-QAM, 64-QAM, and 256-QAM downlink modulation methods may be about 1%, 2%, 3%, and 4%, respectively.

(72) However, the downlink modulation methods provided herein and the second reference downlink BLER value corresponding to each downlink modulation method may only be an example for describing the second condition. In addition, the second condition may include a condition on various downlink modulation methods and a second reference downlink BLER on each downlink modulation method.

(73) The processor **305**, when the second condition is not satisfied, may periodically identify,

through the PUCCH, the downlink BLER in the ET mode and the downlink modulation method in the ET mode and may determine whether the downlink BLER in the ET mode and the downlink modulation method in the ET mode satisfy the second condition. The processor **305**, when the second condition is satisfied, may switch the power supply mode to the ET mode.

(74) FIGS. **4A** and **4B** are diagrams illustrating a condition on a downlink BLER and a downlink modulation method in a power supply control method according to one example embodiment.

(75) FIGS. **4A** and **4B** illustrate an example of switching a power supply mode in the first condition in Table 1 and the second condition in Table 2 provided as examples with reference to FIG. **3**.

(76) Referring to FIG. **4A**, in a case that QPSK, 16-QAM, or 64-QAM downlink modulation method is used in an ET mode, a processor (e.g., the processor **305** of FIG. **3**), when a downlink BLER is greater than or equal to a first reference downlink BLER of about 3%, may determine that the first condition is satisfied and switch the power supply mode from the ET mode to the APT mode.

(77) Referring to FIG. **4B**, in a case that a 256-QAM downlink modulation method is used in the ET mode, the processor **305**, when the downlink BLER is greater than or equal to the first reference downlink BLER of about 5%, may determine that a second condition is satisfied and switch the power supply mode from the ET mode to the APT mode.

(78) The downlink BLER may increase as the number of symbols used in downlink modulation, and when applying the same first reference downlink BLER of about 3% to the QPSK, 16-QAM, 64-QAM, and 256-QAM, the 256-QAM downlink modulation method in the ET mode may frequently exceed the first reference downlink BLER of about 3% corresponding to the downlink BLER in the ET mode. Therefore, a greater value (e.g., 5%) than the first reference downlink BLER of the QPSK, 16-QAM, and 64-QAM downlink modulation methods may be desirable to be the first reference downlink BLER corresponding to the 256-QAM downlink modulation method.

(79) The processor **305** may identify, through a PUCCH, in the APT mode, the downlink BLER in the ET mode and the downlink modulation method in the ET mode and may determine whether the downlink BLER in the ET mode and the downlink modulation method in the ET mode satisfy a second condition.

(80) In FIG. **4A**, in a case that the QPSK, 16-QAM, or 64-QAM downlink modulation method is used in the ET mode, the processor **305**, when the downlink BLER is less than the second reference downlink BLER of about 1%, may determine that the second condition is satisfied and switch the power supply mode from the APT mode to the ET mode.

(81) In FIG. **4B**, in a case that the 256-QAM downlink modulation method is used in the ET mode, the processor **305**, when the downlink BLER is less than the second reference downlink BLER of about 3%, may determine that the second condition is satisfied and switch the power supply mode from the APT mode to the ET mode.

(82) The second reference downlink BLER may be a downlink BLER less than the first reference downlink BLER by a threshold value (e.g., 2% or other designated values). The threshold value may be set to prevent frequent switching between the ET mode and the APT mode. However, the threshold value of about 2% is merely an example. A threshold value between the first reference downlink BLER and the second reference downlink BLER may be set variously.

(83) FIG. **5** is a flowchart illustrating a power supply control method according to one example embodiment.

(84) Referring to FIG. **5**, in operation **505**, a communication apparatus (e.g., the communication apparatus of FIG. **3**) may identify a power supply mode for supplying power, and in operation **510**, may determine whether the power supply mode is an ET mode.

(85) When the power supply mode is not the ET mode in operation **510**, the communication apparatus **300**, in operation **505**, may identify the power supply mode for supplying power again.

(86) In operation **515**, the communication apparatus **300**, when the power supply mode is the ET mode, may determine whether a downlink BLER in the ET mode and a downlink modulation

method in the ET mode satisfy a first condition. The communication apparatus **300**, when the downlink BLER in the ET mode is greater than or equal to a first reference downlink BLER corresponding to the downlink modulation method used in the ET mode, may determine that the first condition is satisfied. The descriptions on the first condition provided with reference to FIG. **3** and FIG. **4A** are not repeated.

(87) When the downlink BLER in the ET mode and the downlink modulation method in the ET mode do not satisfy the first condition in operation **515**, the communication apparatus **300**, in operation **505**, may identify the power supply mode for supplying power again.

(88) When the downlink BLER in the ET mode and the downlink modulation method in the ET mode satisfy the first condition in operation **515**, the communication apparatus **300**, in operation **520**, may determine whether a full RB in an uplink band is allocated to the uplink data transmission.

(89) When the full RB in the uplink band is allocated in operation **520**, the communication apparatus **300** may switch the power supply mode to the APT mode in operation **535**.

(90) When only a partial RB in the uplink band is allocated in operation **520** (i.e., the full RB in the uplink band is not allocated), the communication apparatus **300**, in operation **525**, may identify, through a PUCCH, a downlink BLER in an APT mode and the downlink BLER in the ET mode, and in operation **530**, may determine whether the downlink BLER in the APT mode is less than the downlink BLER in the ET mode.

(91) When the downlink BLER in the APT mode is not less than the downlink BLER in the ET mode in operation **530**, the communication apparatus **300**, in operation **505**, may identify the power supply mode for supplying power again.

(92) When the downlink BLER in the APT mode is less than the downlink BLER in the ET mode in operation **530**, the communication apparatus **300**, in operation **535**, may switch the power supply mode to the APT mode.

(93) The communication apparatus **300**, based on the first condition, whether the full RB in an uplink band is allocated to the uplink data transmission, and/or a result of comparing the downlink BLER in the APT mode to the downlink BLER in the ET mode, may determine whether to switch from the ET mode to the APT mode. Accordingly, the communication apparatus **300** may reduce the number of times switching to the APT mode and additional power consumed by using the APT mode.

(94) To determine whether to switch the power supply mode from the APT mode to the ET mode, the communication apparatus **300**, in operation **540**, may identify the downlink BLER in the ET mode and the downlink modulation method in the ET mode through the PUCCH, and in operation **545**, may determine whether the downlink BLER in the ET mode and the downlink modulation method in the ET mode satisfy a second condition. The communication apparatus **300**, when the downlink BLER in the ET mode is less than a second reference downlink BLER corresponding to the downlink modulation method used in the ET mode, may determine that the second condition is satisfied. The descriptions on the second condition provided with reference to FIG. **3** and FIG. **4B** are not repeated.

(95) When the second condition is not satisfied in operation **545**, the communication apparatus **300**, in operation **540**, may identify, through the PUCCH, the downlink BLER in the ET mode and the downlink modulation method in the ET mode again.

(96) When the second condition is satisfied in operation **545**, the communication apparatus **300**, in operation **550** may switch the power supply mode to the ET mode.

(97) FIG. **6** is a flowchart illustrating a power supply control method according to one example embodiment.

(98) Referring to FIG. **6**, the power supply control method may include: determining whether a power supply mode for supplying power from a power modulator **315** of a communication apparatus **300** to an amplifier **320** for amplifying an RF signal is an ET mode for supplying the

power based on the ET of the RF signal in operation **605**; when the power supply mode is the ET mode, determining whether a downlink BLER in the ET mode and a downlink modulation method in the ET mode satisfy a first condition in operation **610**; when the first condition is satisfied, determining whether a full RB in an uplink band is allocated to uplink data transmission in operation **615**; and when the full RB in the uplink band is allocated to the uplink data transmission, switching the power supply mode to an APT mode for supplying power based on average power of the RF signal in operation **620**.

(99) Operation **605** may include operations **505** and **510** of FIG. 5. Operation **610** may correspond to operation **515** of FIG. 5. Operation **615** may correspond to operation **520** of FIG. 5. Operation **620** may correspond to operation **535** of FIG. 5.

(100) The power supply control method, when a partial RB in the uplink band is allocated to uplink data transmission, may further include: comparing a downlink BLER in the APT mode with the downlink BLER in the ET mode; and when the downlink BLER in the APT mode is less than the downlink BLER in the ET mode, switching the power supply mode to the APT mode.

(101) The power supply control method may further include: determining, in the APT mode, whether the downlink BLER in the ET mode and the downlink modulation method in the ET mode satisfy a second condition; and when the second condition is satisfied, switching the power supply mode to the ET mode.

(102) The first condition may include a condition whether the downlink BLER in the ET mode is greater than or equal to a first reference downlink BLER corresponding to the downlink modulation method in the ET mode, and the determining whether the first condition is satisfied may include determining that the first condition is satisfied when the downlink BLER in the ET mode is greater than or equal to the first reference downlink BLER corresponding to the downlink modulation method in the ET mode.

(103) The second condition may include a condition on whether the downlink BLER in the ET mode is less than a second reference downlink BLER corresponding to the downlink modulation method in the ET mode, and the determining whether the second condition is satisfied may include determining that the second condition is satisfied when the downlink BLER in the ET mode is less than the second reference downlink BLER corresponding to the downlink modulation method in the ET mode.

(104) The second reference downlink BLER may be a downlink BLER less than the first reference downlink BLER by a threshold value.

(105) The comparing the downlink block error rate in the APT mode with the downlink block error rate in the ET mode may be performed through a PUCCH.

(106) The RF signal may include data transmitted through a PUSCH.

(107) The first reference downlink BLER corresponding to the downlink modulation method in the ET mode in the first condition may have a greater value as more symbols are used in the downlink modulation method in the ET mode.

(108) A communication apparatus **300** may include an amplifier **320** for amplifying an RF signal, a power modulator **315** for supplying power to the amplifier **320**, and a processor **305** for controlling the power modulator **315**. The processor **305** may determine whether a power supply mode for supplying the power from the power modulator **315** to the amplifier **320** is an ET mode for supplying the power based on ET of the RF signal, when the power supply mode is the ET mode, determine whether a downlink BLER in the ET mode and a downlink modulation method in the ET mode satisfy a first condition, when the first condition is satisfied, determine whether a full RB in an uplink band is allocated to uplink data transmission, and when the full RB in the uplink band is allocated to the uplink data transmission, switch the power supply mode to an APT mode for supplying the power based on average power of the RF signal.

(109) The processor **305**, when a partial RB in the uplink band is allocated to uplink data transmission, may compare a downlink BLER in the APT mode with the downlink BLER in the ET

mode, and when the downlink BLER in the APT mode is less than the downlink BLER in the ET mode, switch the power supply mode to the APT mode.

(110) The processor **305**, may determine, in the APT mode, whether the downlink BLER in the ET mode and the downlink modulation method in the ET mode satisfy the second condition, and when the second condition is satisfied, switch the power supply mode to the ET mode.

(111) The first condition may include a condition on whether the downlink BLER in the ET mode is greater than or equal to a first reference downlink BLER corresponding to the downlink modulation method in the ET mode, and the processor **305** may determine that the first condition is satisfied when the downlink BLER in the ET mode is greater than or equal to the first reference downlink BLER corresponding to the downlink modulation method in the ET mode.

(112) The second condition may include a condition on whether the downlink BLER in the ET mode is less than a second reference downlink BLER corresponding to the downlink modulation method in the ET mode, and the processor **305** may determine that the second condition is satisfied when the downlink BLER in the ET mode is less than the second reference downlink BLER corresponding to the downlink modulation method in the ET mode.

(113) The second reference downlink BLER may be a downlink BLER less than the first reference downlink BLER by a threshold value.

(114) The processor **305** may compare the downlink BLER in the APT mode with the downlink BLER in the ET mode through a PUCCH.

(115) The RF signal may include data transmitted through a PUSCH.

(116) The first reference downlink BLER corresponding to the downlink modulation method in the ET mode in the first condition may have a greater value as more symbols are used in the downlink modulation method in the ET mode.

(117) The processor **305**, when the second condition is not satisfied, in the APT mode, periodically perform the determining whether the downlink BLER in the ET mode and the downlink modulation method in the ET mode satisfy the second condition.

(118) The electronic device according to various example embodiments may be one of various types of electronic devices. The electronic device may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance device. According to an example embodiment of the disclosure, the electronic device is not limited to those described above.

(119) It should be understood that various example embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. In connection with the description of the drawings, like reference numerals may be used for similar or related components. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things unless the relevant context clearly indicates otherwise. As used herein, “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B or C”, “at least one of A, B and C”, and “A, B, or C,” each of which may include any one of the items listed together in the corresponding one of the phrases, or all possible combinations thereof. Terms such as “first”, “second”, or “first” or “second” may simply be used to distinguish the component from other components in question, and do not limit the components in other aspects (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., by wire), wirelessly, or via a third element.

(120) As used in connection with various example embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may

interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an example embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(121) Various example embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., the internal memory **136** or the external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium and execute it. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include code generated by a compiler or code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Here, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

(122) According to an example embodiment, a method according to various example embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., a compact disc read-only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smartphones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as a memory of the manufacturer's server, a server of the application store, or a relay server.

(123) According to various example embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components.

(124) According to various example embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various example embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various example embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

Claims

1. A communication apparatus comprising: an amplifier for amplifying a radio frequency (RF) signal; a power modulator for supplying power to the amplifier; and a processor for controlling the power modulator, wherein the processor is configured to determine whether a power supply mode for supplying the power from the power modulator to the amplifier is an envelope tracking (ET) mode for supplying the power based on ET of the RF signal, when the power supply mode is the ET mode, determine whether a downlink block error rate in the ET mode and a downlink modulation method in the ET mode satisfy a first condition, when the first condition is satisfied, determine whether a full resource block (RB) in an uplink band is allocated to uplink data

transmission, and when the full RB in the uplink band is allocated to the uplink data transmission, switch the power supply mode to an average power tracking (APT) mode for supplying the power based on average power of the RF signal.

2. The communication apparatus of claim 1, wherein the processor is configured to, when a partial RB, not the full RB, in the uplink band is allocated to the uplink data transmission, compare a downlink block error rate in the APT mode with the downlink block error rate in the ET mode, and when the downlink block error rate in the APT mode is less than the downlink block error rate in the ET mode, switch the power supply mode to the APT mode.

3. The communication apparatus of claim 2, wherein the processor is configured to, in the APT mode, determine whether the downlink block error rate in the ET mode and the downlink modulation method in the ET mode satisfy a second condition, and when the second condition is satisfied, switch the power supply mode to the ET mode.

4. The communication apparatus of claim 3, wherein the processor is configured to, when the second condition is not satisfied, in the APT mode, periodically perform the determining whether the downlink block error rate in the ET mode and the downlink modulation method in the ET mode satisfy the second condition.

5. The communication apparatus of claim 3, wherein the first condition comprises: a condition whether the downlink block error rate in the ET mode is greater than or equal to a first reference downlink block error rate corresponding to the downlink modulation method in the ET mode, and the processor is configured to, when the downlink block error rate in the ET mode is greater than or equal to the first reference downlink block error rate corresponding to the downlink modulation method in the ET mode, determine that the first condition is satisfied.

6. The communication apparatus of claim 5, wherein, in the first condition, the first reference downlink block error rate corresponding to the downlink modulation method in the ET mode has a greater value as a total number of symbols used in the downlink modulation method in the ET mode increases.

7. The communication apparatus of claim 5, wherein the second condition comprises: a condition whether the downlink block error rate in the ET mode is less than a second reference downlink block error rate corresponding to the downlink modulation method in the ET mode, and the processor is configured to, when the downlink block error rate in the ET mode is less than the second reference downlink block error rate corresponding to the downlink modulation method in the ET mode, determine that the second condition is satisfied.

8. The communication apparatus of claim 7, wherein the second reference downlink block error rate is a downlink block error rate less than the first reference downlink block error rate by a threshold value.

9. The communication apparatus of claim 2, wherein the processor is configured to compare the downlink block error rate in the APT mode with the downlink block error rate in the ET mode through a physical uplink control channel (PUCCH).

10. The communication apparatus of claim 1, wherein the RF signal comprises: data transmitted through a physical uplink shared channel (PUSCH).

11. A method of controlling power supply for amplifying a radio frequency (RF) signal, the method comprising: determining whether a power supply mode for supplying power from a power modulator of a communication apparatus to an amplifier for amplifying the RF signal is an envelope tracking (ET) mode for supplying the power based on ET of the RF signal; when the power supply mode is the ET mode, determining whether a downlink block error rate in the ET mode and a downlink modulation method in the ET mode satisfy a first condition; when the first condition is satisfied, determining whether a full resource block (RB) in an uplink band is allocated to uplink data transmission; and when the full RB in the uplink band is allocated to the uplink data transmission, switching the power supply mode to an average power tracking (APT) mode for supplying the power based on average power of the RF signal.

12. The method of claim 11, further comprising: when a partial RB, not the full RB, in the uplink band is allocated to the uplink data transmission, comparing a downlink block error rate in the APT mode with the downlink block error rate in the ET mode, and when the downlink block error rate in the APT mode is less than the downlink block error rate in the ET mode, switching the power supply mode to the APT mode.
 13. The method of claim 12, wherein the comparing the downlink block error rate in the APT mode with the downlink block error rate in the ET mode is performed through a physical uplink control channel (PUCCH).
 14. The method of claim 12, further comprising: in the APT mode, determining whether the downlink block error rate in the ET mode and the downlink modulation method in the ET mode satisfy a second condition, and when the second condition is satisfied, switching the power supply mode to the ET mode.
 15. The method of claim 14, wherein the first condition comprises: a condition whether the downlink block error rate in the ET mode is greater than or equal to a first reference downlink block error rate corresponding to the downlink modulation method in the ET mode, and determining whether the first condition is satisfied comprises: when the downlink block error rate in the ET mode is greater than or equal to the first reference downlink block error rate corresponding to the downlink modulation method in the ET mode, determining that the first condition is satisfied.
 16. The method of claim 15, wherein, in the first condition, the first reference downlink block error rate corresponding to the downlink modulation method in the ET mode has a greater value as a total number of symbols used in the downlink modulation method in the ET mode increases.
 17. The method of claim 15, wherein the second condition comprises: a condition whether the downlink block error rate in the ET mode is less than a second reference downlink block error rate corresponding to the downlink modulation method in the ET mode, and determining whether the second condition is satisfied comprises: when the downlink block error rate in the ET mode is less than the second reference downlink block error rate corresponding to the downlink modulation method in the ET mode, determining that the second condition is satisfied, wherein the second reference downlink block error rate is a downlink block error rate less than the first reference downlink block error rate by a threshold value.
 18. The method of claim 11, wherein the RF signal comprises: data transmitted through a physical uplink shared channel (PUSCH).
 19. A non-transitory computer-readable storage medium storing instructions that, when executed by a processor, cause the processor to perform the method of claim 11.
 20. A method of controlling power supply for amplifying a RF signal, the method comprising: determining whether a power supply mode for supplying power from a power modulator of a communication apparatus to an amplifier for amplifying the RF signal is an ET mode for supplying the power based on ET of the RF signal; when the power supply mode is the ET mode, determining whether a downlink block error rate in the ET mode and a downlink modulation method in the ET mode satisfy a first condition; when the first condition is satisfied, determining whether a partial RB in an uplink band is allocated to uplink data transmission; when the partial RB in the uplink band is allocated to the uplink data transmission, comparing a downlink block error rate in an APT mode with the downlink block error rate in the ET mode, and when the downlink block error rate in the APT mode is less than the downlink block error rate in the ET mode, switching the power supply mode to the APT mode.
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